# THE DIAGONAL OF THE DERIVED-COMPLEXES STACK IS AFFINE OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 2221-2225 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 2227-2263, the complete author proof (35 body lines).

*Editorial note.* Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 10 of queue_moduli.json (pb=1520, stmt 2221-2225, proof 2227-2263).

*Editorial note.* stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 0DLG: sources/stacks-moduli/pinned-original-source.tex, source lines 2263–2263. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0DLC: sources/stacks-moduli/pinned-original-source.tex, source lines 2263–2263. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 08K9: sources/stacks-moduli/pinned-original-source.tex, source lines 2263–2263. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *The diagonal of $\mathcal{C}omplexes_{X/B}$ over $B$ is affine and of finite presentation.*

*Proof.* The representability of the diagonal by algebraic spaces was shown in Quot, Lemma 0DLG. From the proof we find that we have to show: given a scheme $T$ over $B$ and objects $E, E' \in D(\mathcal{O}_{X_T})$ such that $(T, E)$ and $(T, E')$ are objects of the

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fibre category of $\mathcal{C}omplexes_{X/B}$ over $T$, then $Isom(E, E') \to T$ is affine and of finite presentation. Here $Isom(E, E')$ is the functor

$$(Sch/T)^{opp} \to Sets, \quad T' \mapsto \{\varphi : E_{T'} \to E'_{T'} \text{ isomorphism in } D(\mathcal{O}_{X_{T'}})\}$$

where $E_{T'}$ and $E'_{T'}$ are the derived pullbacks of $E$ and $E'$ to $X_{T'}$. Consider the functor $H = \mathcal{H}om(E, E')$ defined by the rule

$$(Sch/T)^{opp} \to Sets, \quad T' \mapsto \mathrm{Hom}_{\mathcal{O}_{X_{T'}}}(E_T, E'_T)$$

By Quot, Lemma 0DLC this is an algebraic space affine and of finite presentation over $T$. The same is true for $H' = \mathcal{H}om(E', E)$, $I = \mathcal{H}om(E, E)$, and $I' = \mathcal{H}om(E', E')$. Therefore we see that

$$Isom(E, E') = (H' \times_T H) \times_{c, I \times_T I', \sigma} T$$

where $c(\varphi', \varphi) = (\varphi \circ \varphi', \varphi' \circ \varphi)$ and $\sigma = (\mathrm{id}, \mathrm{id})$ (compare with the proof of Quot, Proposition 08K9). Thus $Isom(E, E')$ is affine over $T$ as a fibre product of schemes affine over $T$. Similarly, $Isom(E, E')$ is of finite presentation over $T$. $\square$